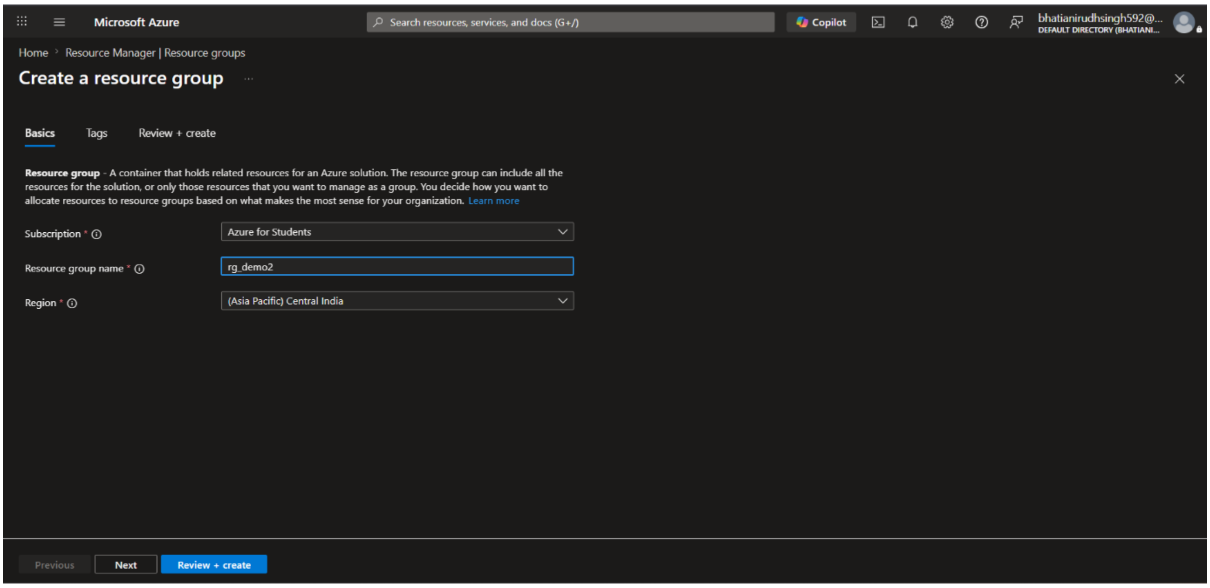
Assignment of Week 4

Step 1: Create Resource Group

Logged in to the Azure Portal and created a new Resource Group named rg-demo2 in the selected Azure region. The Resource Group acts as a logical container for managing all Azure resources used in this project.

Creating...



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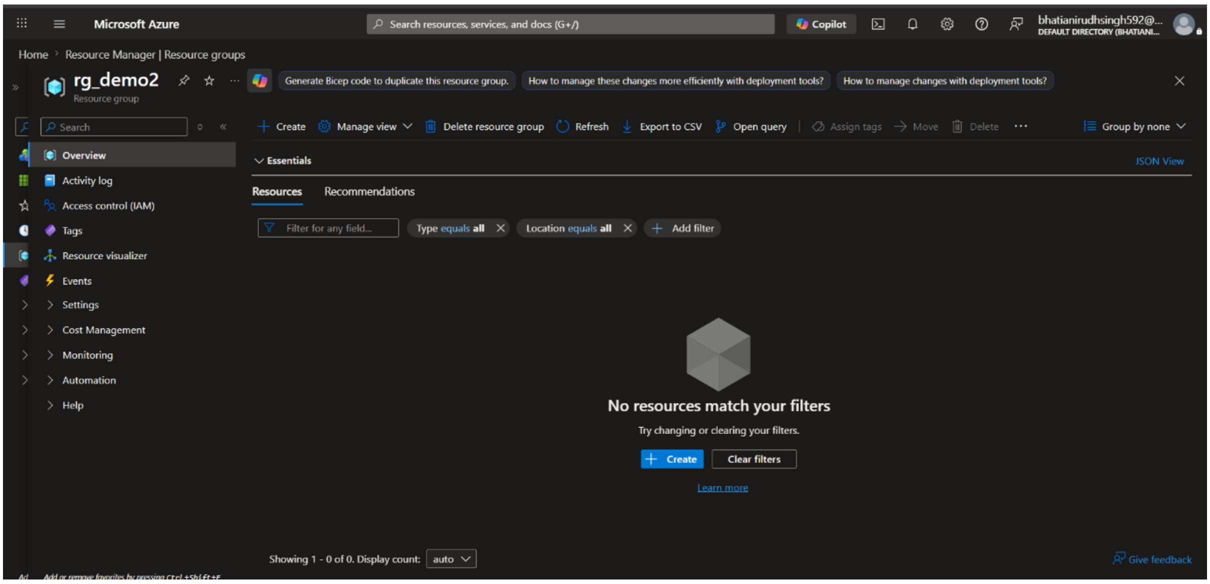
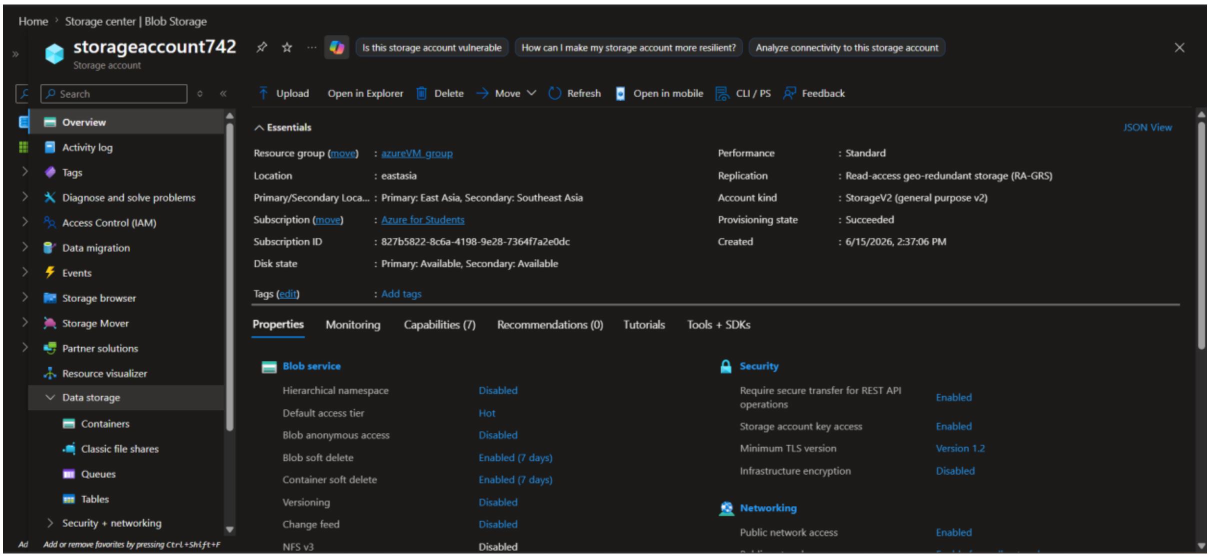


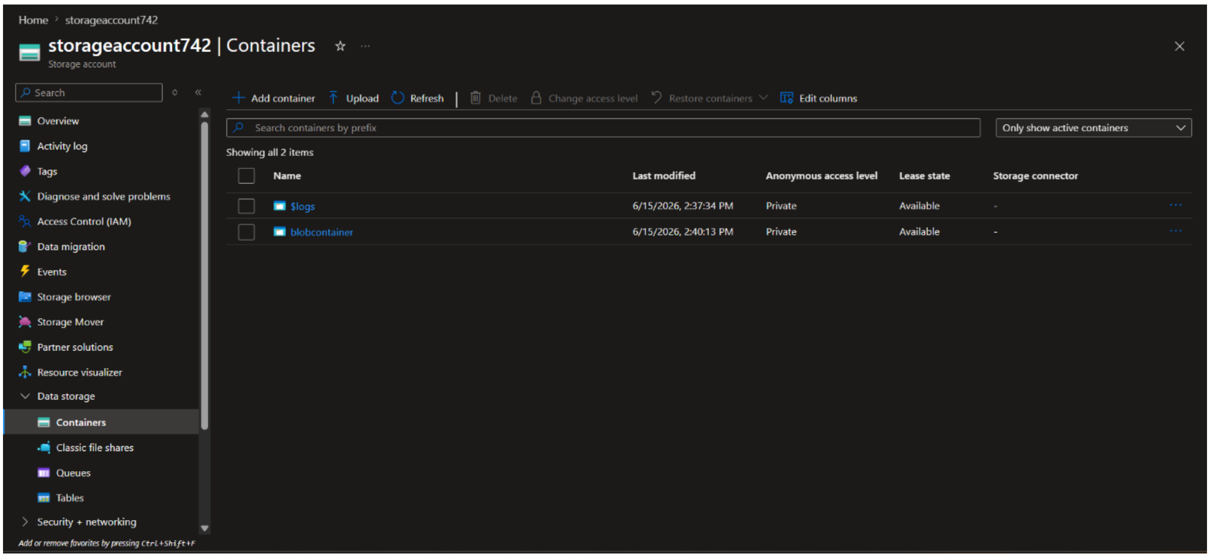
Step 2: Create Storage Account

Created an Azure Storage Account named **storageaccount742** to store source and destination data files. The storage account provides scalable and secure cloud storage services.



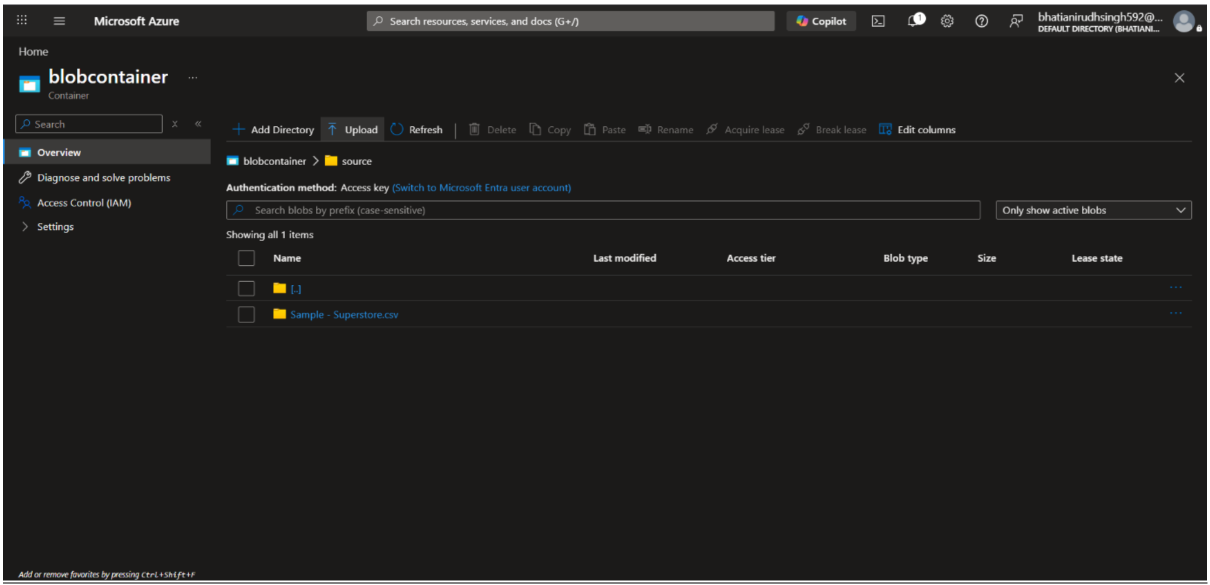
Step 3: Create Blob Containers

Created Blob Containers inside the Storage Account:



Step 4: Upload Source CSV File

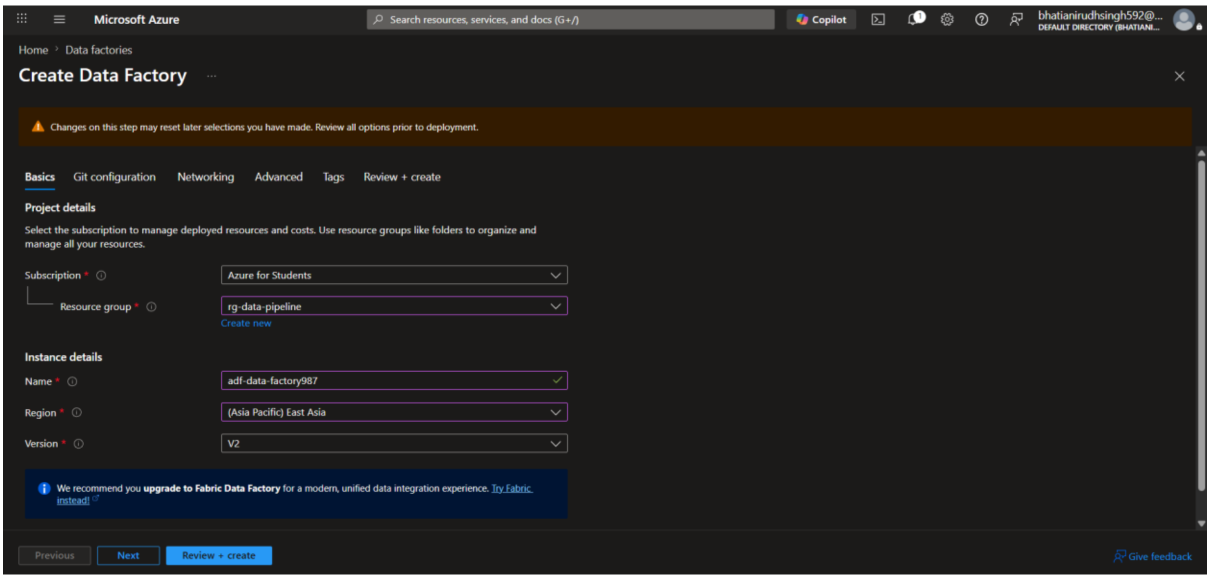
Uploaded the sample CSV file (Sample_Superstore.csv) into the source container. This file will be used as the source dataset for the pipeline.



Step 5: Create Azure Data Factory

Created an Azure Data Factory instance named adf-data-factory987 and launched Azure Data Factory Studio for pipeline development and monitoring.

Creating...



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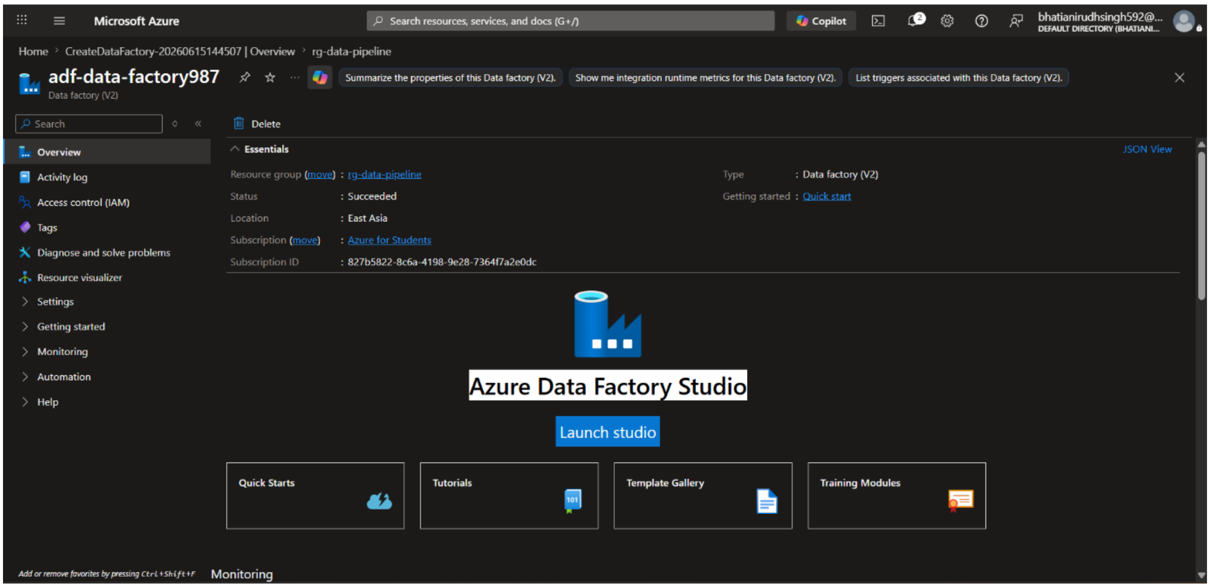
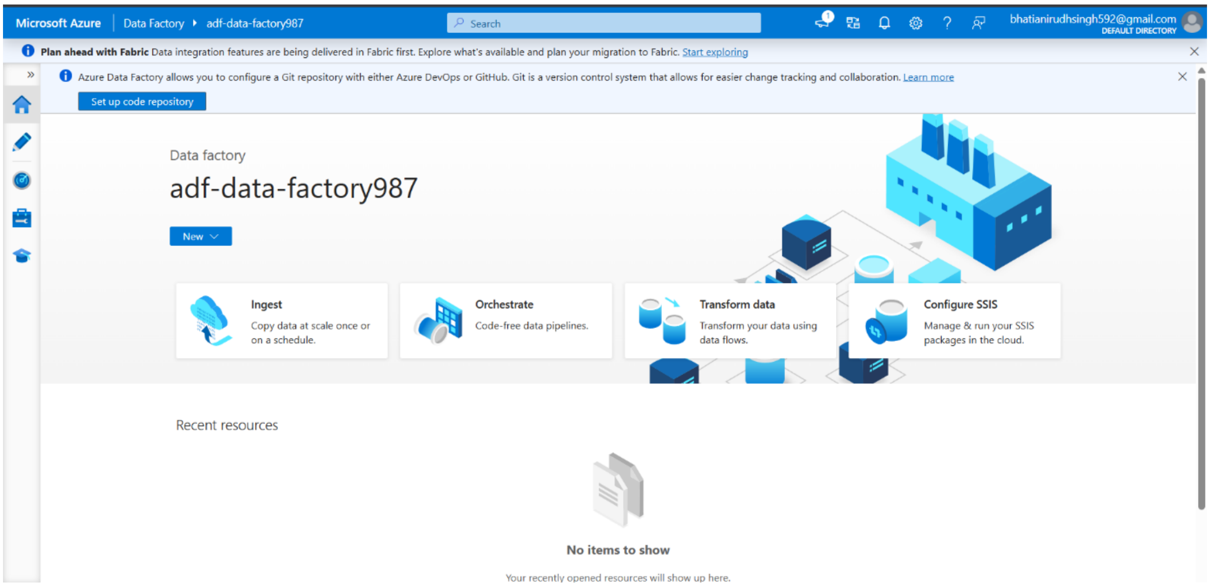


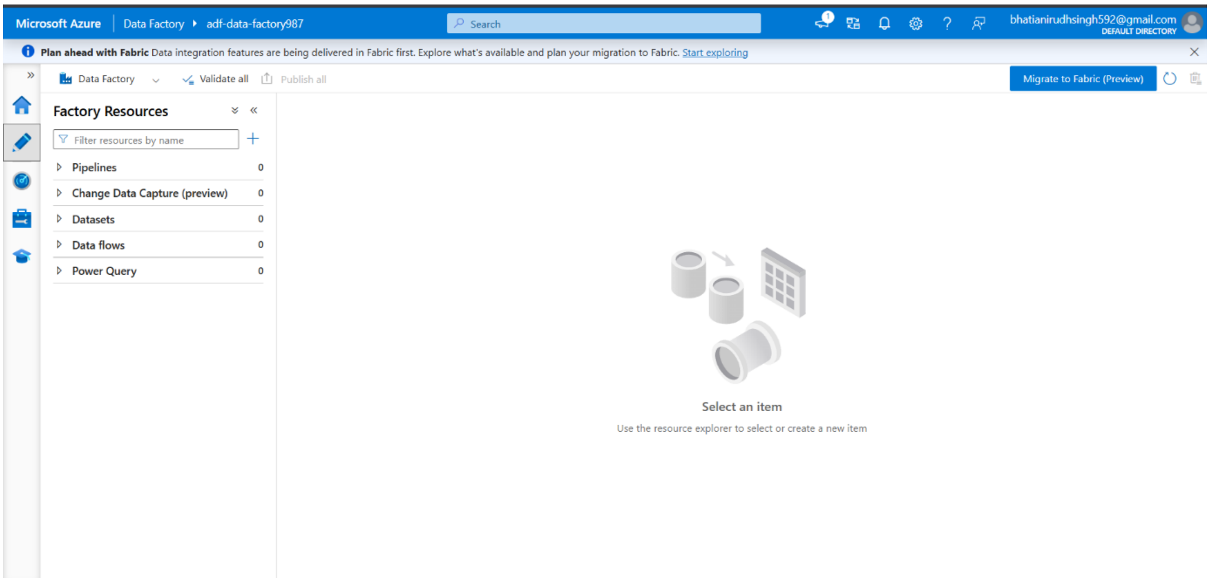
Launched



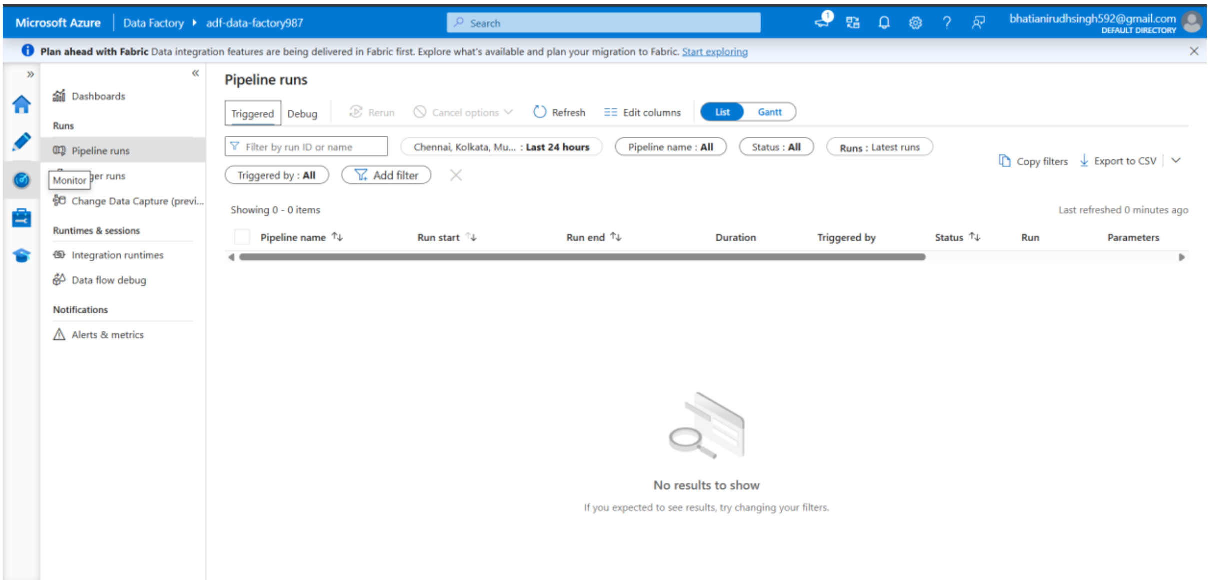
Step 6: Explore Azure Data Factory Studio

Explored the main sections of Azure Data Factory Studio:

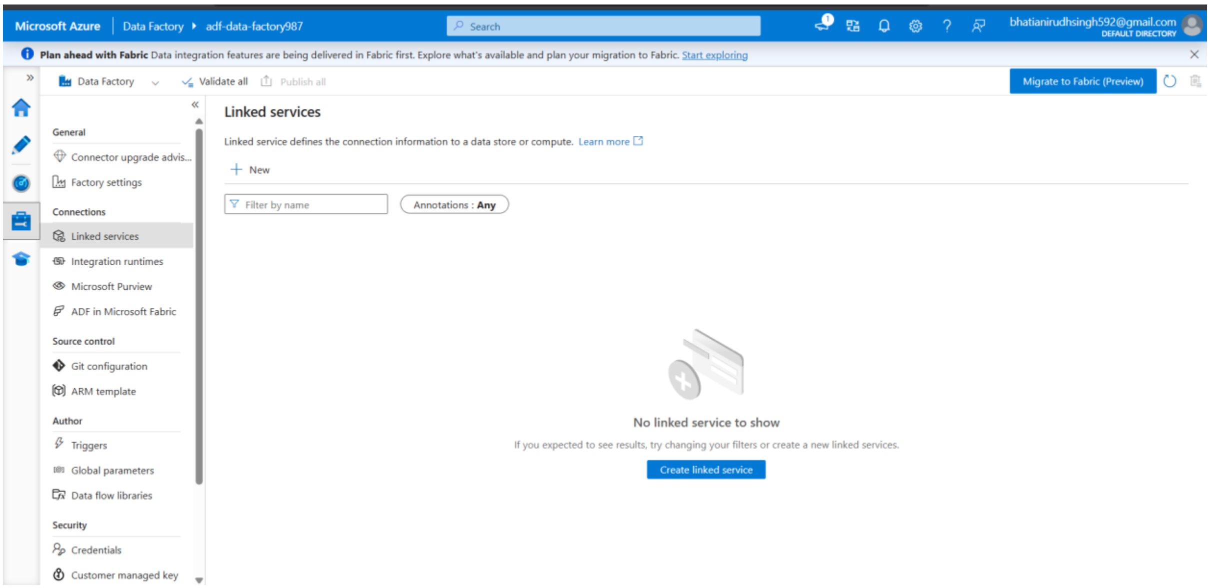
Author



Monitor

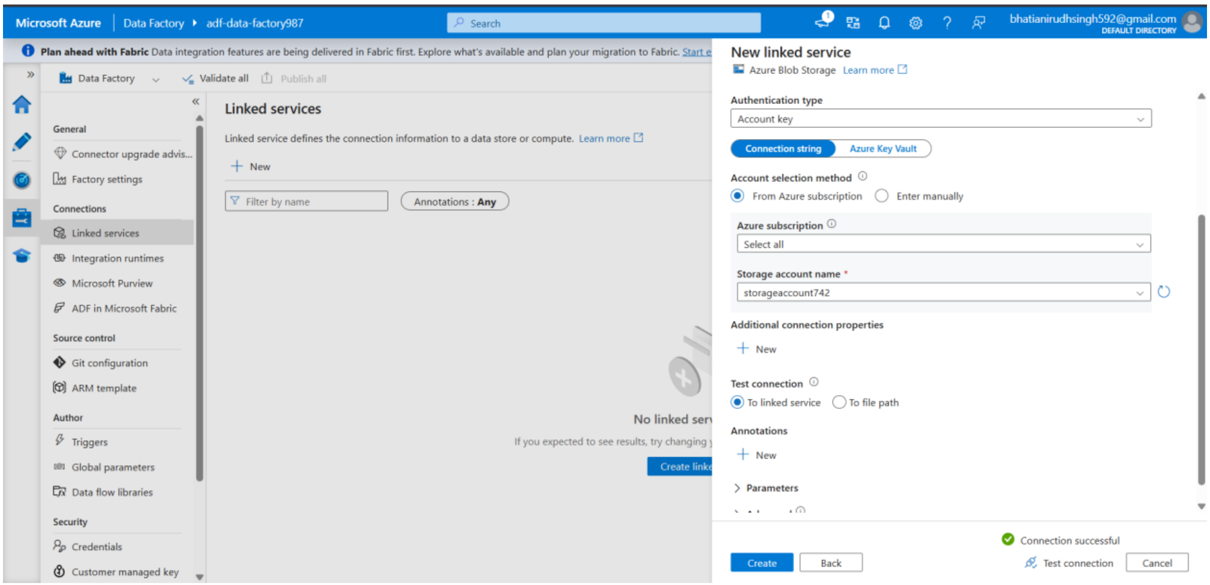


Manage



Step 7: Create Linked Service

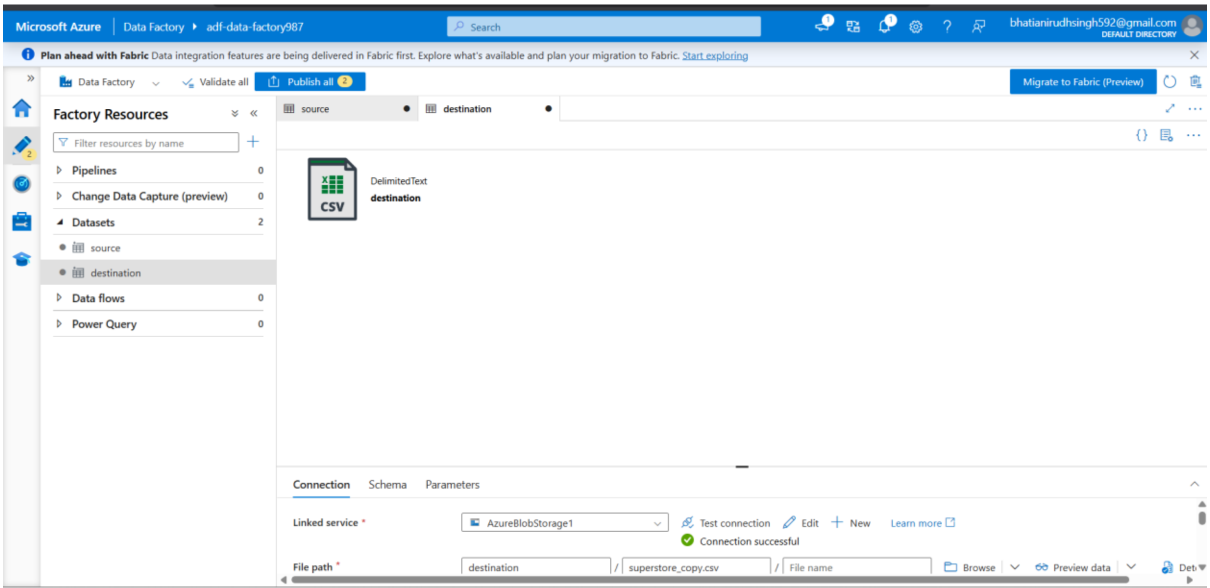
Configured a Linked Service connecting Azure Data Factory with Azure Blob Storage. This connection allows ADF to access files stored in the Storage Account.



Step 8: Create Source and Destination Dataset

Created a source dataset pointing to the Sample_Superstore.csv file located in the source container. Imported the schema to validate file structure.

Created a destination dataset pointing to the destination container where the copied file will be stored after pipeline execution.



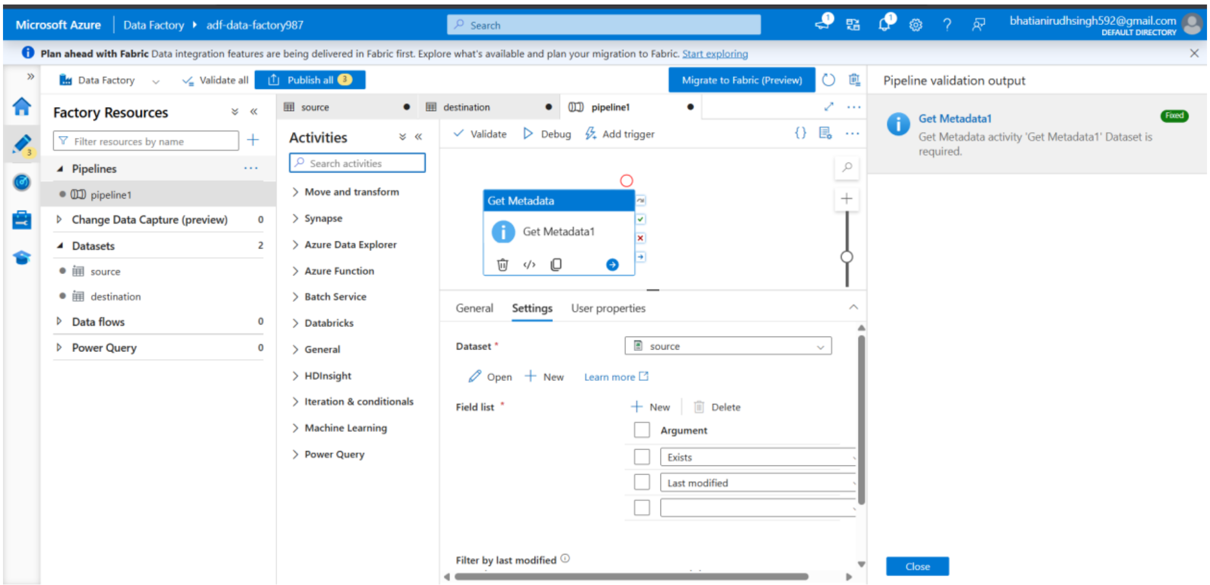
Step 9: Create Pipeline & Configure Get Metadata Activity

Created a new pipeline named **pipeline** to orchestrate metadata validation and file movement activities.

Added a Get Metadata activity to retrieve source file information including:

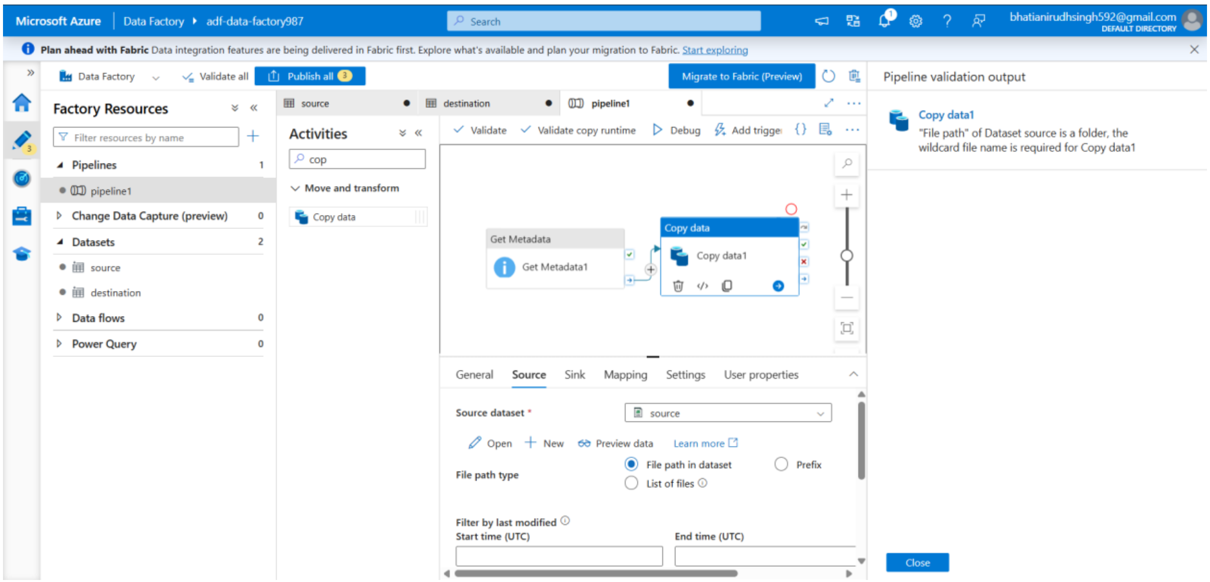
- File existence
- Last modified timestamp

This validates the source file before data movement.



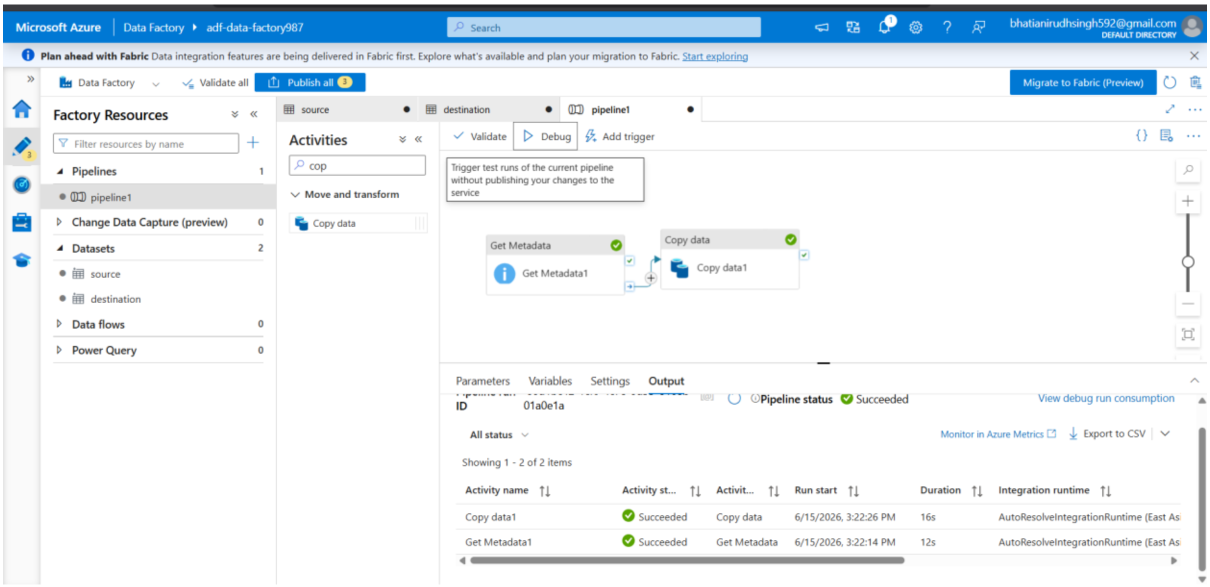
Step 10: Configure Copy Data Activity

Added a Copy Data activity to copy the CSV file from the source container to the destination container.



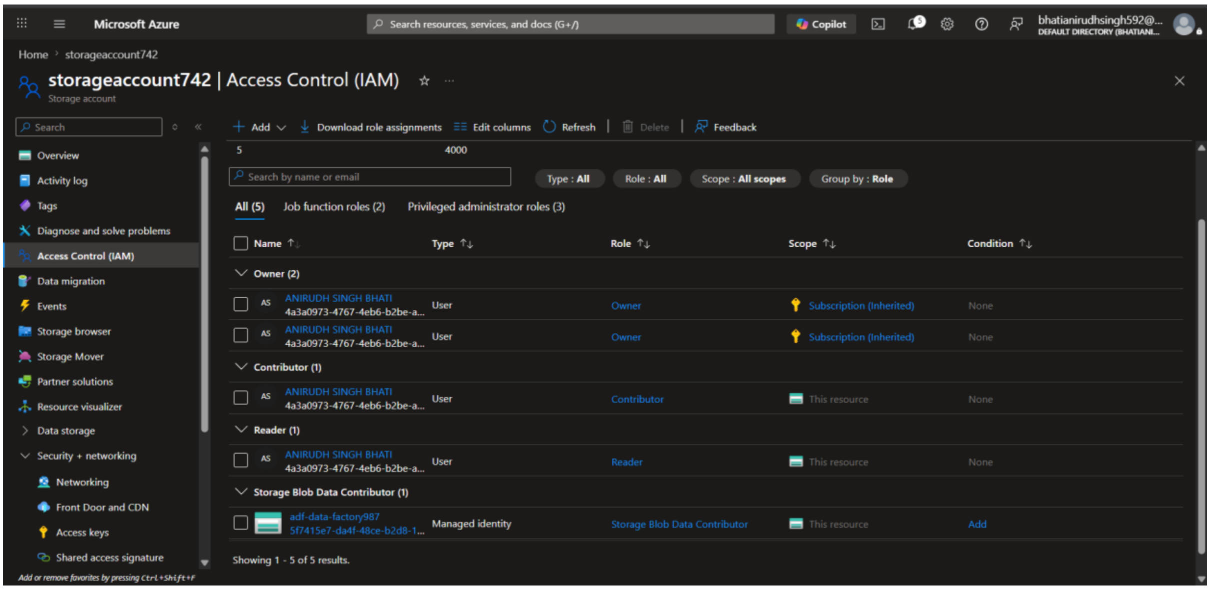
Step 11: Validate Pipeline

Validated the pipeline configuration to ensure there were no errors before execution.



Step 13: Configure IAM Permissions

Assigned appropriate permissions between Azure Data Factory and Azure Storage using IAM role assignments to enable secure access.



Step 14: Execute Pipeline Using Debug

Executed the pipeline using the Debug option to test functionality before production execution.